

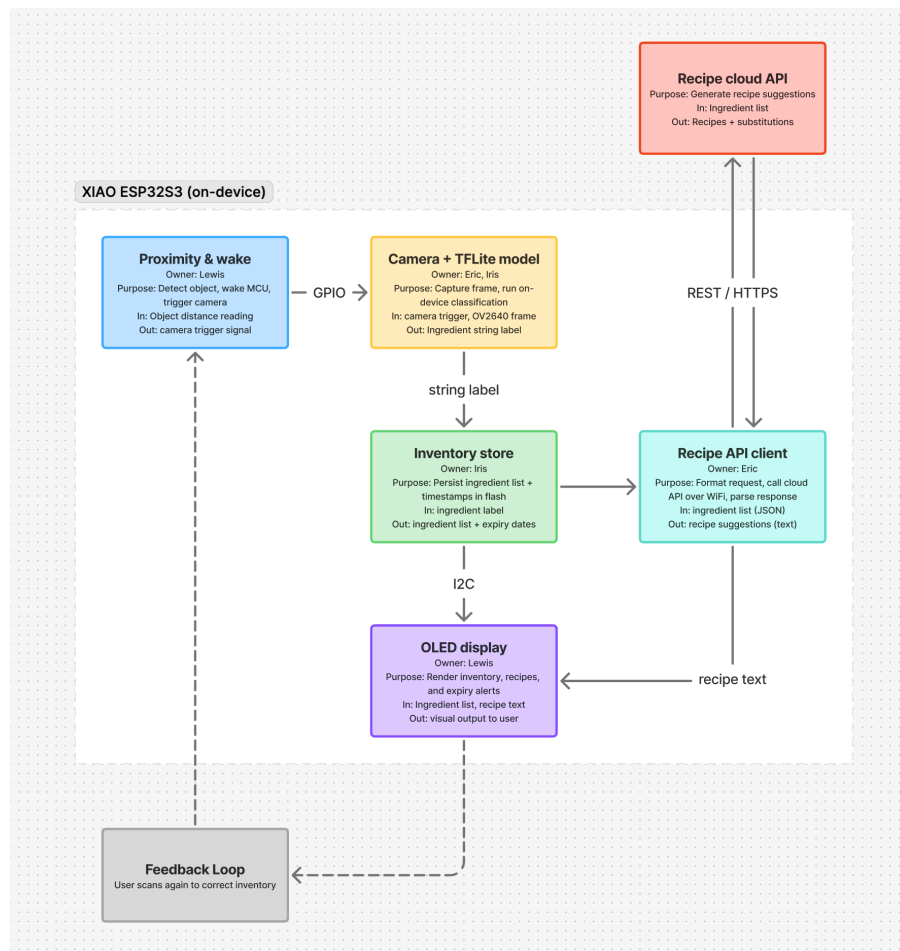
Project Register and Ownership

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Section 1: System Mission & Boundary

- FridgeLens is a magnet-mounted smart inventory device that helps home cooks track fridge contents and reduce food waste.
- It is designed for people who cook at home regularly but struggle to keep track of what is in their fridge.
- Our system runs on a XIAO ESP32S3 with an OV2640 camera module, a proximity sensor, and an OLED display. The basic concept will be the proximity sensor detects a receipt or ingredient being held up to the device, it automatically wakes the camera and identifies the item. After that, all the recognized ingredients will be added to or removed from a stored inventory list in flash memory. When the User is taking out an ingredient from the fridge, the OLED screen will show the expiry reminders along with recipe suggestions.

Section 2: Functional Architecture Diagram



Section 3: Risk Reduction Prototypes

Field	Machine Learning - Food
Name + Owner	Food-item image classification feasibility Owner: Eric
Assumption being validated	A MobileNet V2 INT8 model running on ESP32-S3 can distinguish at least 10 common food ingredients with $\geq 80\%$ accuracy, at ≤ 500 ms per inference.
Inputs -> Process -> Outputs	Inputs: 200+ labeled images of 10 ingredient classes. Process: Train MobileNet V2, deploy to ESP32-S3, run inference on 40 held-out images. Outputs: Confusion matrix, per-class accuracy, average inference latency (ms).

Field	Machine Learning - Receipt
Name + Owner	Receipt OCR parsing feasibility Owner: Iris
Assumption being validated	A cloud OCR API can extract a structured ingredient list from a photographed grocery receipt with $\geq 85\%$ item-level accuracy across varied receipt formats and lighting conditions.
Inputs -> Process -> Outputs	Inputs: 30+ receipt photos taken under varied lighting and angles. Process: Send receipt images to cloud OCR API, parse returned text into ingredient list. Outputs: Item-level extraction accuracy (%), failure cases annotated by error type.

Field	Enclosure + PCB assemble
Name + Owner	Enclosure of our sensor and display device Owner: Iris, Lewis
Assumption being validated	3D print the enclosure of our project and assemble the surfboard or the custom PCB.
Inputs -> Process -> Outputs	Outputs: Go/no-go fit assessment, functional perfboard assembly, and annotated photos of any clearance issues or required design revisions.

Field	Hardware
Name + Owner	Distance-sensor-triggered camera capture pipeline Owner: Lewis

Assumption being validated	The HC-SR04 sensor can reliably detect an object approaching within 10 cm and trigger a clean camera capture within 300 ms, with < 5% false-trigger rate
Inputs -> Process -> Outputs	Inputs: HC-SR04 mounted at fixed distance from camera FOV; OV2640 connected to ESP32-S3. Process: Object approaches sensor 50 times (varied speeds/angles). HC-SR04 reads distance, triggers capture when threshold crossed. Outputs: Trigger success rate (%), sample captured images for visual inspection.

Field	Backend
Name + Owner	End-to-end cloud recipe generation round-trip Owner: Eric
Assumption being validated	The ESP32-S3 can POST a JSON ingredient list over WiFi to a cloud endpoint, receive a structured recipe response, and render it on a 128×64 OLED in under 5 seconds total round-trip time.
Inputs -> Process -> Outputs	Inputs: Food List ["tomato", "egg", "scallion"]. Process: ESP32-S3 connects to WiFi, sends HTTPS POST, cloud generates recipe via LLM, returns structured JSON. Outputs: Round-trip latency, successful render on OLED, recipe quality.